

Know

Trace Minerals – Guided Notes – NDFS 1020

IRON									
Roles • hemoglobin & myoglobin • Formation of new cells • Cognition	Deficiency • Anemia – symptoms include: • fatigue • Impaired cognition • Cold hands • pale skin • Dizziness • Shortness of breath • Deficiency is common among • pregnant women (increased blood volume) • Menstruating women (monthly blood loss) • People with parasites (GI bleeding)	Marker in Blood <table border="1"> <tr> <td>Plasma ferritin</td> <td>Iron storage</td> </tr> <tr> <td>Transferrin</td> <td>Iron transport</td> </tr> <tr> <td>Transferrin Saturation</td> <td>How full the iron transporters are</td> </tr> </table>	Plasma ferritin	Iron storage	Transferrin	Iron transport	Transferrin Saturation	How full the iron transporters are	What it Measures
Plasma ferritin	Iron storage								
Transferrin	Iron transport								
Transferrin Saturation	How full the iron transporters are								
T/F: If a person has normal red blood cells (typical size and color), they have sufficient iron in their body. <u>False</u>		Hemochromatosis is an iron-storage disease caused by inadequate or insufficient amounts of a hormone called <u>hepcidin</u> .							
Low ferritin and <u>transferrin</u> saturation are indicators of iron depletion and iron deficiency (precursors to iron deficiency anemia).		Symptoms include: • Organ dysfunction (liver, heart, etc.) • Fatigue, depression • Increased <u>bacterial</u> infection							
Which of the following is best absorbed and utilized in the body? (circle one) • Heme iron (found in meat, fish, poultry) • Non-heme iron (found in plants)		RDA • Men and post-menopausal women – <u>8</u> mg • Pre-menopausal women – <u>18</u> mg • Vegetarians RDA x <u>1.8</u>							
Factors that Increase Iron Absorption/Status	Factors that Decrease Iron Absorption								
<u>Vitamin C</u> Cooking food in a cast iron skillet	<u>Tannins</u> <u>Calcium</u>								
ZINC									
Roles: • Growth and reproduction • <u>taste</u> perception • *High doses may reduce <u>HDL</u> levels	Deficiency: • Impaired <u>growth</u> and maturation • Iron-deficiency <u>anemia</u>	Food sources: • Protein foods • Seafood • Oysters • Meats, fish, poultry • Whole grains							
IODINE									
Roles: • Part of <u>thyroid</u> hormones • Contributes to metabolism • <u>Goiter</u> • Deficiency AND toxicity • Cretinism • Deficiency	Food sources: • <u>iodized</u> salt • Produce (depends on soil)								
FLUORIDE									
Roles: Aids in formation of <u>bones</u> and <u>teeth</u>		Too much: <u>Fluorosis</u> (discoloration and weakness of teeth)							
Deficiency: dental caries; bone abnormalities		Source: Fluoridated water							

What is the most common nutrient deficiency? Iron

Which mineral is included in enrichment of processed grains? Iron

Iron

- Functions
 - Energy metabolism
 - Growth
 - Reproduction

- Wound healing
- Immune function

TABLE 12.2 Roles of Iron in the Body

Body Function or System	Role
Oxygen transport and energy metabolism	Needed for production of hemoglobin (red blood cells), myoglobin (muscle cells), and cytochromes (most body cells)
Cell division	Required by an enzyme needed for DNA production
Immune system	Needed for production of lymphocytes (a type of white blood cell) Enables neutrophils (another type of white blood cell) to destroy bacteria
Nervous system	Needed to help maintain the myelin sheath that covers parts of certain nerve cells Needed for the production of neurotransmitters (e.g., dopamine, epinephrine, and norepinephrine) that regulate brain and muscular activity

- Food sources
 - Heme: Beef, fish, poultry
 - Nonheme: meat, veggies, grains, supplements, fortified or enriched foods
 - A mixed diet will result in absorption of 14-18%, a plant diet is 5-12%
 - The iron added to grains is not very bioavailable

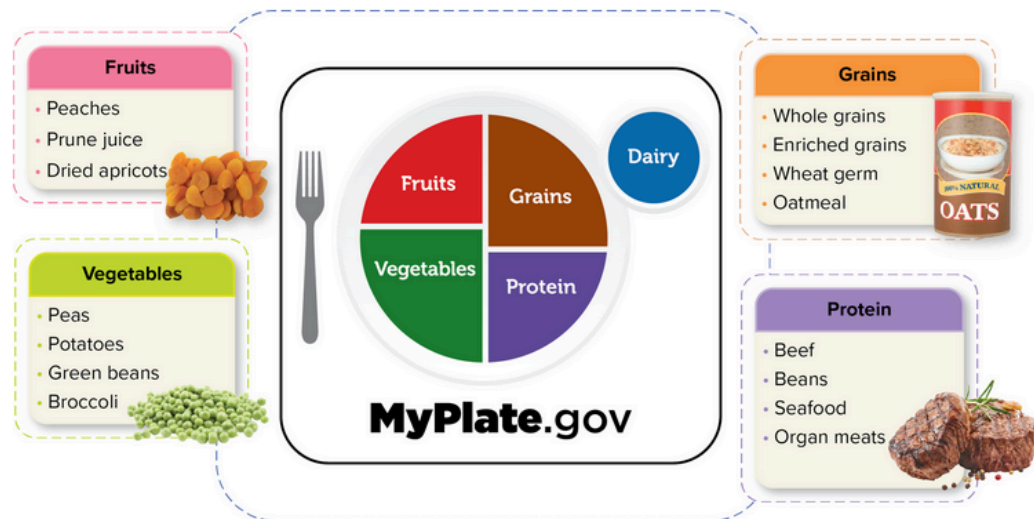
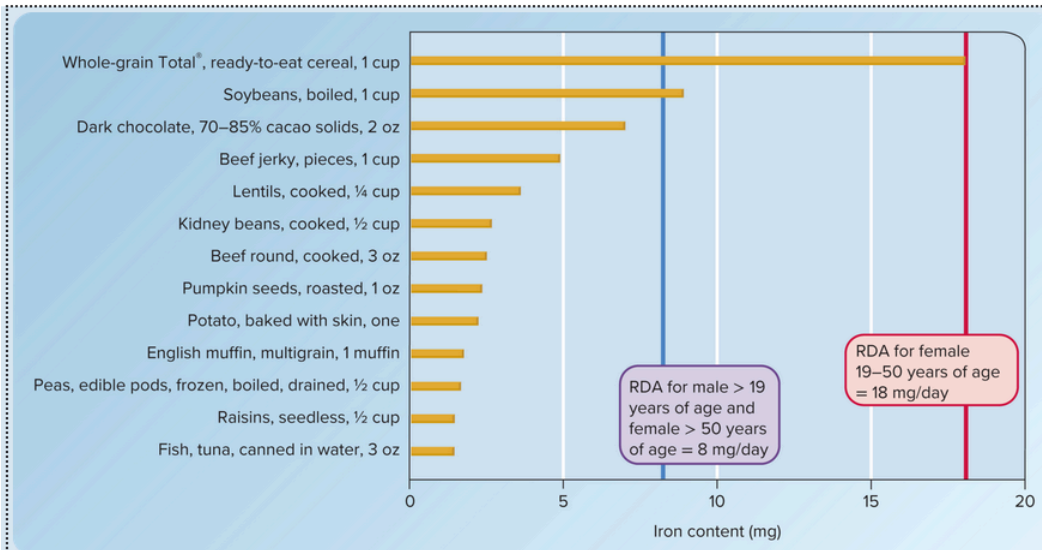


TABLE 12.3 Iron Content of Selected Foods



- Dietary adequacy
 - RDA = 8 mg for males and post menopausal women, 18 mg for females. Vegetarians have a x1.8 modifier
 - Deficiency
 - Stage 1: iron depletion
 - No apparent health problems, drop in ferritin level
 - Stage 2: iron deficiency
 - Reduced ferritin level
 - Elevated level of transferrin receptors
 - Stage 3: iron deficiency anemia
 - Small, pale red blood cells
 - Very high level of transferrin receptors
 - Symptoms of iron deficiency, poor pregnancy outcomes, heart failure
 - Interferes with growth, behavior, immune system function, energy metabolism, development of cognitive functioning in children

TABLE 12.5 Signs and Symptoms of Iron Deficiency Anemia

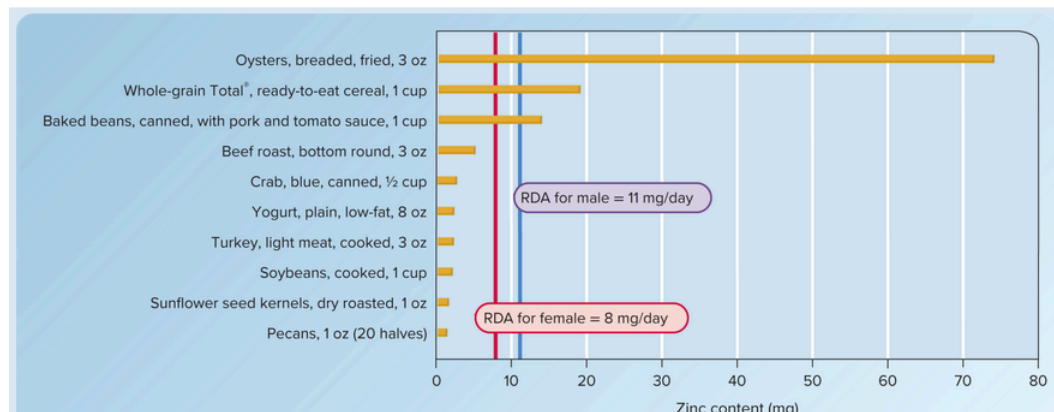
- Fatigue and weakness
- Shortness of breath
- Dizziness
- Headache
- Coldness in the hands and feet
- Pale skin
- Chest pain

- Over time can cause rapid or irregular heartbeat, chest pain, enlarged heart, and heart failure
- Toxicity
 - UL = 45 mg/day
 - Excess iron in blood acts as an oxidant/free radical
 - Accumulation in tissues
 - Joint pain
 - Abnormal bronze skin color
 - liver/heart/adrenal glands/pancreas damage
 - Hereditary hemochromatosis
 - Fatigue
 - Lack of energy
 - Abdominal pain
 - Loss of sex drive
 - Heart problems
 - Organ damage

Zinc

- Functions
 - Functions of 50+ enzymes
 - Growth and development
 - Wound healing
 - Sense of taste and smell
 - DNA synthesis
 - Proper functioning of nervous and immune systems
- Food sources
 - Red meat, poultry

TABLE 12.7 Zinc Content of Selected Foods



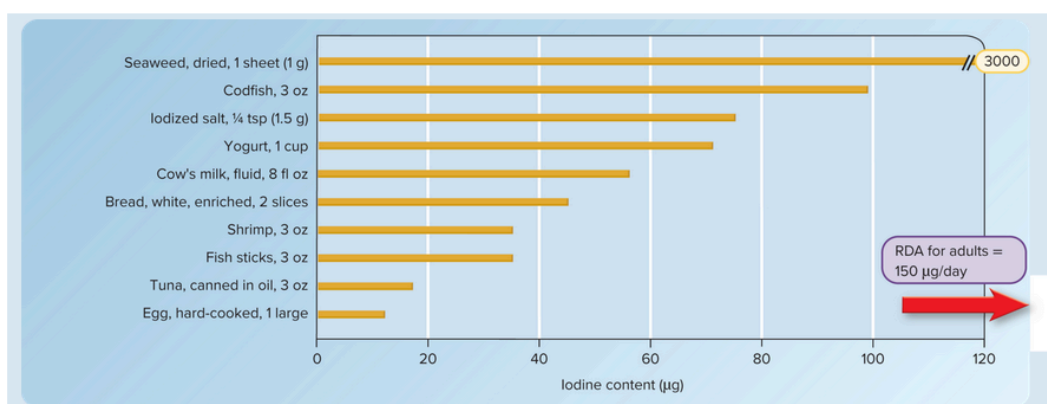
- Bioavailability increases when eaten with proteins, especially with sulphur, and when not a lot of other metals are competing for absorption
- Dietary adequacy
 - RDA = 8 - 11 mg

- Plant based diet = x1.5
- Deficiency
 - Alcoholics, chronic diarrhea, digestive tract issues, older adults at risk
 - diarrhea
 - Loss of appetite
 - Hair loss
 - Skin rash
 - Poor wound healing
 - Impaired sense of taste
 - Intellectual disabilities
 - Weakened immune system
 - Severe cases
 - Stunted growth in children leading to dwarfism and failure of sexual maturation
 - During pregnancy
 - Preterm birth
 - Low birth weight
- Toxicity
 - UL = 40 mg
 - Reduces beneficial HDL levels
 - Diarrhea
 - Cramps
 - Nausea
 - Vomiting
 - Depressed immune system function
 - Interferes with copper absorption and metabolism

Iodine

- Functions
 - Thyroid function and production of thyroid hormones
 - Controls rate of cellular metabolism
- Food sources
 - Seaweed, saltwater fish, grains, cows milk, eggs, iodized salt

TABLE 12.6 Iodine Content of Selected Foods

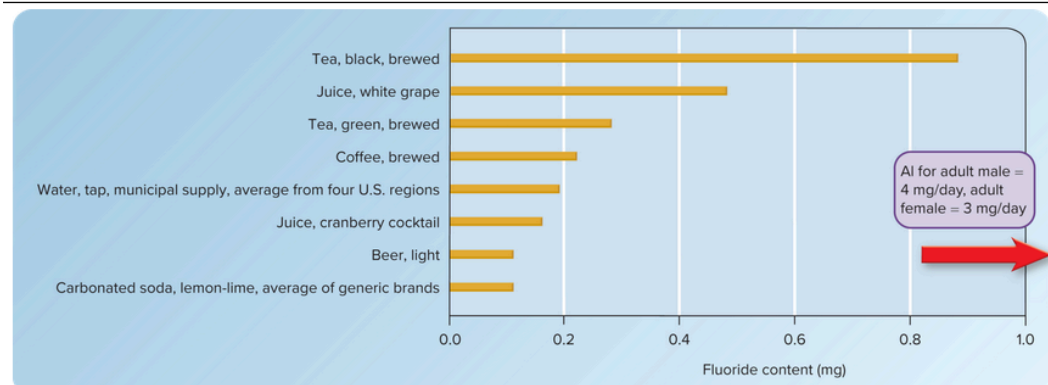


- Dietary adequacy
 - RDA = 150 micrograms
 - Deficiency
 - Requirements increase during pregnancy and lactation
 - Deficiency can result in low birth weight, stillbirths, abnormal fetal brain development (congenital hypothyroidism)
 - Hyperthyroidism
 - Low blood levels of thyroid hormone
 - Reduced metabolic rate
 - Elevated blood cholesterol
 - Fatigue
 - Difficulty concentrating on mental tasks
 - Weight gain
 - Constipation
 - Dry skin
- Toxicity
 - UL = 1.1 mg
 - Goiter
 - Hypothyroidism OR hyperthyroidism
 - Increased risk of thyroid cancer

Fluoride

- Functions
 - Remineralize tooth surface and prevent cavities from forming
 - In kids, build teeth resistant to decay
 - Promote bone development
 - Incorporates into bone tissue
- Food sources
 - Fluoridated water, coffee, tea, soft drinks, beer, and juice
 - Fish, shellfish, crab, shrimp, clams

TABLE 12.9 Fluoride Content of Selected Beverages (8 fl oz)



- Dietary adequacy
 - AI = 4 mg for males, 3mg for females
 - Deficiency
 - negligible
- Toxicity
 - Dental fluorosis (abnormal appearance of tooth enamel; white specks, severe discoloration, pitting)
 - Over long periods of time, can lead to skeletal fluorosis which causes joint stiffness and bone pain
 - Increased risk of fractures

Skim

What is a trace mineral

- Body needs under 100 mg /day

TABLE 12.1 Summary of Trace Minerals

Mineral	Major Functions in the Body	Adult RDA/AI* (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Symptoms
Iron (Fe)	• Component of hemoglobin and myoglobin that carries oxygen • Energy generation • Immune system function	Female: 18 mg Male: 8 mg	Meat and other animal foods, except cow's milk; whole-grain and enriched breads and cereals; fortified cereals	• Fatigue on exertion • Small, pale red blood cells • Low hemoglobin levels • Poor immune system function • Developmental disabilities in infants	Adult Upper Intake Level (UL) = 45 mg/day • Intestinal upset • Organ damage • Death
Zinc (Zn)	• Component of numerous enzymes	Female: 8 mg Male: 11 mg	Seafood, meat, whole grains	• Skin rash • Diarrhea • Depressed sense of taste and smell • Hair loss • Poor growth and physical development	UL = 40 mg/day • Intestinal upset • Depressed immune system function • Supplement use can reduce copper absorption.
Copper (Cu)	• Promotes iron metabolism • Component of antioxidant enzymes • Component of enzymes involved in connective tissue synthesis	0.9 mg	Liver, cocoa, legumes, whole grains, shellfish	• Anemia • Reduced immune system function • Poor growth and development	UL = 10 mg/day • Vomiting • Abnormal nervous system function • Liver damage

Selenium (Se)	• Component of an antioxidant system	55 μg	Meat, eggs, fish, seafood, whole grains	• Muscle pain and weakness • Form of heart disease	UL = 400 $\mu\text{g}/\text{day}$ • Nausea • Vomiting • Hair loss • Weakness • Liver damage
Iodine (I)	• Component of thyroid hormones	150 μg	Iodized salt, saltwater fish, dairy products	• Goiter • Congenital hypothyroidism	UL = 1100 $\mu\text{g}/\text{day}$ • Reduced thyroid gland function
Fluoride (F)	• Increases resistance of tooth enamel to cavity formation • Stimulates bone formation	Female: 3 mg Male: 4 mg	Fluoridated water, tea, seaweed	• No true deficiency, but increased risk of tooth decay	UL = 10 mg/day • Stomach upset • Staining of teeth during development • Bone deterioration
Chromium (Cr)	• Enhances insulin action	Female: 20–25 μg Male: 30–35 μg	Egg yolks, whole grains, pork, nuts, mushrooms	• Blood glucose level remains elevated after meals.	• Unknown but currently under scientific investigation • May interact with certain medications
Manganese (Mn)	• Cofactor for certain enzymes, including some involved in carbohydrate metabolism	Female: 1.8 mg Male: 2.3 mg	Nuts, oats, beans, tea	• None in healthy humans	UL = 11 mg/day • Abnormal nervous system function
Molybdenum (Mo)	• Component of certain coenzymes	45 μg	Liver, peas, beans, cereal products, leafy vegetables, cow's milk	• None in healthy humans	UL = 2000 $\mu\text{g}/\text{day}$ • Rarely occurs from usual dietary sources • Overdoses of dietary supplements containing molybdenum may cause joint pain and swelling of feet or lower legs.

Iron: absorption, storage, and transport

- Iron is used to make ferritin (major storage form)
- Stored in liver, spleen, bone marrow
- If body has low iron, ferritin levels drop which signals to small intestine to absorb more
- Transferrin = protein in blood that transports iron. Binds to a receptor on cell membrane to deliver iron to cell
 - Levels in blood increase when iron is low
- Intestinal tract absorbs ferrous iron more efficiently than the ferric form
 - Ferric $\rightarrow$ ferrous, in acidic conditions
 - Oxalic acid in plants reduces absorption

TABLE 12.4 Factors That Influence Iron Bioavailability

Enhance	Inhibit
High intake of vitamin C with iron-containing food	High intake of calcium with iron-containing food
Consuming more heme, as compared to nonheme, forms of iron	Medications that reduce stomach acidity
Consuming bread that has been leavened	Oxalic acid from foods such as spinach
Consuming iron-containing foods that are fermented	Phytic acid from foods such as whole grains
Soaking of beans or grains before consuming	Soy protein from foods such as tofu
	Polyphenols from foods and beverages such as tea

Iron deficiency, high risk populations

- 10-19% of females age 12-49
- Causes
 - Substantial blood loss (injuries, heavy periods)
 - Increased iron requirements during rapid growth or pregnancy (increased blood volume)
- Plant based diets also increase risk due to different levels of bioavailability

Iodine: absorption

- Body absorbs 85-90% of iodine present in food
- Some plants(cassava, turnips, cabbage, brussel sprouts, cauliflower, broccoli, soy products) contain goitronins that inhibit iodide metabolism and reduces thyroid hormone production
 - Not a problem unless you eat a lot routinely

Fluoride: fluoridation of public water supplies

- About 75% of US population has access
- 0.7-1.2 mg/L, safely prevents dental caries
- Over lifetime, can reduce likelihood of tooth decay by 25%
- Filtered or bottled water does not contain fluoride
- Some ppl don't like it because they think it is involuntary medication or whatever